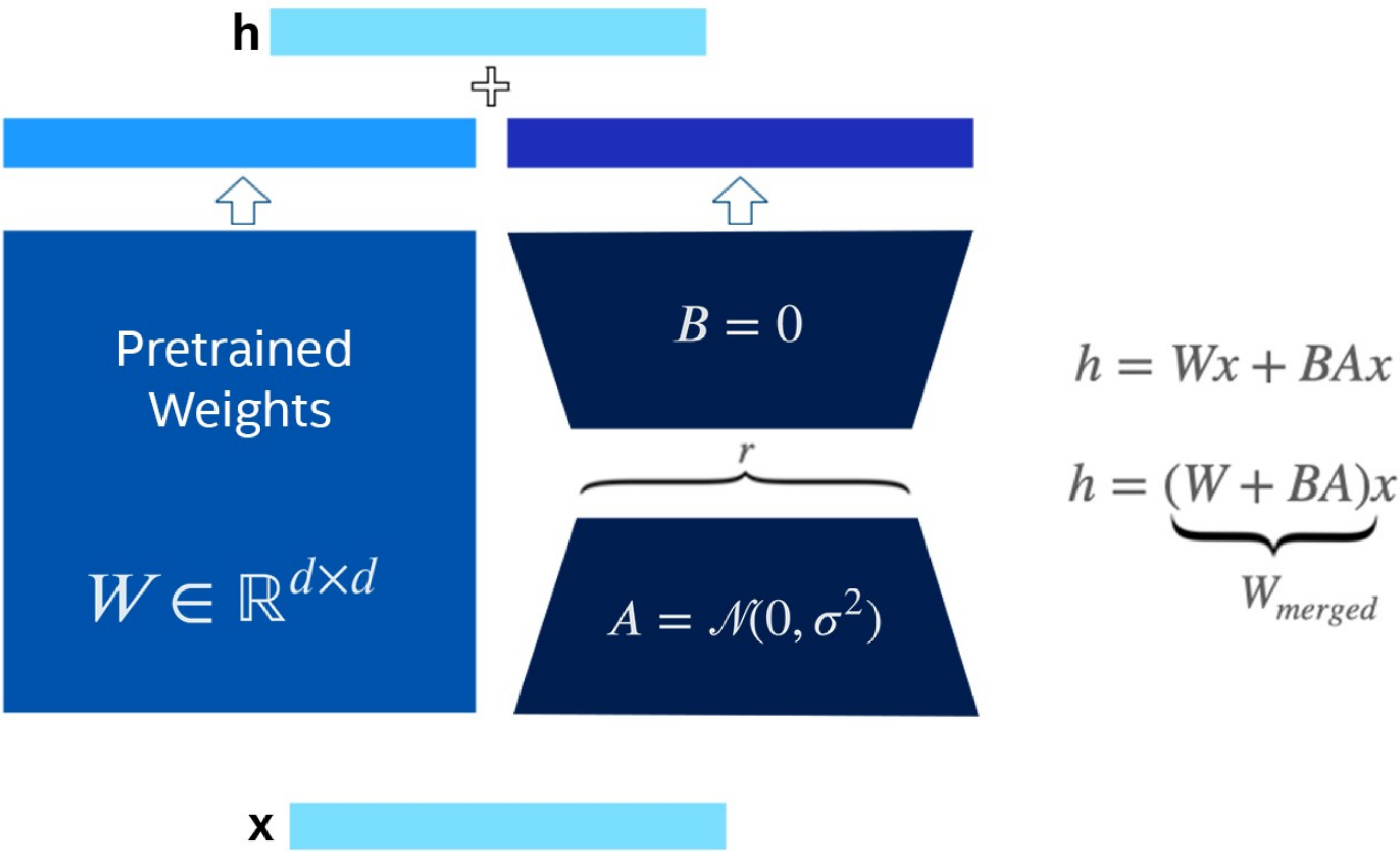
Parameter Efficient Fine-Tuning (PEFT)



https://www.intel.cn/content/www/cn/zh/developer/articles/training/fine-tuning-llama2-model-with-lora.html

Initialization strategies

$$\text{Xavier: } W_{i,j} \sim U\left(-\sqrt{\frac{1}{f_{in}}}, \sqrt{\frac{1}{f_{in}}}\right)$$

$$\text{He: } W_{i,j} \sim U\left(-\sqrt{\frac{2}{f_{in}}}, \sqrt{\frac{2}{f_{in}}}\right)$$

$$\text{Glorot: } W_{i,j} \sim U\left(-\sqrt{\frac{6}{f_{in}+f_{out}}}, \sqrt{\frac{6}{f_{in}+f_{out}}}\right)$$

- Most strategies:
  - Are simple and heuristic
  - Try to achieve nice properties during initialization
  - We don't know if the nice properties are preserved during training
  - We don't know how initialization affects generalization
- Some tips:
  - We don't want weights with the same values
    - Random initialization using a Gaussian or a uniform distribution
  - Large values may lead to exploding or vanishing gradients

Documentation: https://docs.pytorch.org/docs/2.12/nn.init.html